

Anwesh Pandey, Ph.D.

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Personal Details

Date of Birth: 10th September 1992

Gender: Male

Place of Birth: Gorakhpur, U.P.

Marital Status: Married

Citizenship: Indian

Languages Known: English, and Hindi

Profile Summary

I'm a motivated researcher passionate about teaching physics & research at intersection of computational chemistry, biology, computer-aided drug discovery, and in-house database development. During my Ph.D., I explored Protein/DNA-ligand underlying noncovalent interactions. More recently, I've been expanding my work into developing manually curated database for enzyme catalytic activity prediction based on sequence information combining bioinformatics with statistical learning approaches. I'm particularly excited about using AI/ML to accelerate drug discovery pipelines, binding affinity prediction, and molecular property optimization, followed by developing databases. I am willing to join an institution where I can scale as an independent researcher and contribute to advancing research, mentoring students and establishing collaborations.

Academic & Research Background

Sep 2016 – Dec 2020 Ph.D. in Applied Physics

Babasaheb Bhimrao Ambedkar University, Lucknow, U.P.

Major: Computational Biophysics, Molecular Modelling, Non-covalent Interactions, and Automation of work flow through bash scripting.

Dissertation: A Computational Study of Drug-DNA Interactions.

PI: Dr. Anil Kumar Yadav

2014 – 2016

Master of Science in Applied Physics

Babasaheb Bhimrao Ambedkar University, Lucknow, U.P.

Major: Molecular Modelling, C-Programming, Laser & Fiber Optics.

Dissertation: A DFT study of electronic properties of FDA approved drugs used in the treatment of respiratory diseases.

PI: Prof. Devesh Kumar

CGPA: 7.58/10

2011 – 2014

Bachelor of Science in Physics & Mathematics

DAV PG College, DDU Gorakhpur University, Gorakhpur, U.P.

Major: Quantum Mechanics, Atomic & Molecular Physics, Vector Algebra, Differential Equations.

Marks (%): 69/100

Work Experiences

July 2026 – Present

Postdoctoral Fellow

Department of Computational Biology, Indraprastha Institute of Information Technology, New Delhi, India.

Area: AI/ML in Ligand Based Drug Design; Database Development; Noncovalent Interactions.

PI: *Dr. N. Arul Murugan*

Oct 2023 – July 2026

Postdoctoral Fellow

School of Pharmacy, The Hebrew University of Jerusalem, Israel.

Area: AI/ML in Enzyme Catalytic Activity Prediction; Database Development; Protein Electrostatics; Noncovalent Interactions; Valence Bond Modelling.

PI: *Prof. Avital Shurki*

Feb 2021 – Jul 2023

CSIR-Senior Project Associate

Advanced Computation and Data Sciences Division, CSIR-North East Institute of Science and Technology, Jorhat, Assam.

Area: Manifestation, and quantification of various Non-Covalent Interactions in complex chemical, biological and supramolecular systems.

PI: *Prof. G. Narahari Sastry*

Jun 2022 – Feb 2023

NOVARTIS – CSIR Industry Collaboration Project

Advanced Computation and Data Sciences Division, CSIR-North East Institute of Science and Technology, Jorhat, Assam.

Area: Studies involving drug delivery via molecular cavitands, viz., cyclodextrins, cucurbiturils, resorcinarenes, etc.; electronic structure and molecular dynamics of cavitand-drug complexes; empirical data collection, manual curation, and pre-processing for the development of ML model to predict binding affinity of cavitand-drug complexes.

PI: *Prof. G. Narahari Sastry*

Sep 2020 – Dec 2020

DST-SERB (Senior Research Fellow)

Special Center for Molecular Medicine, Jawaharlal Nehru University, New Delhi.

Topic: Comparison of *E. coli* and *S. aureus* in antimicrobial resistance in Human DNA topoisomerase 1A using advanced computational tools, Protein-Ligand Interactions for development of novel COVID therapeutics.

PI: *Prof. Vibha Tandon*

Feb 2020 – Mar 2020

Internship

Department of Chemistry, Indian Institute of Technology, Guwahati, Assam.

Area: Shell Scripting, Python & C Programming, Automation of work flow in computational chemistry & biology softwares.

PI: *(Late) Dr. Hemant Kumar Srivastava*

Research Interests

- **Computational Quantum Chemistry:** Functionalization studies of carbanous nanostructures; interactions between host-guest complexes and encapsulated host-guest complexes; studies of prevalence and size dependence on functionalization and encapsulation; and benchmark calculations.
- **Non-Covalent Interactions:** Characterization, quantification, manifestations and prevalence of different non-covalent interactions manifesting in complex chemical, biological and supramolecular systems; classification studies through quantum theory of atoms in molecules; manifestations of cooperativity; and quantification of interactions through energy decomposition analysis.
- **Computer Aided Drug Design:** Modelling structure and dynamics of complex biological macromolecules via screening, docking, molecular dynamics simulations and free energy calculations; in-depth analysis of the interface interactions between DNA-ligand, protein-ligand, protein-DNA, protein-DNA-ligand, and protein-protein complexes; incorporation of high-end QM & QM/MM methods for detailed studies of the underlying interactions.
- **Database Development:** Developing a manually curated database for haloalkane dehalogenase catalytic activity prediction; integrating protein sequence information through Direct Coupling Analysis (DCA); including empirical catalytic activity, sequence information; data curation, data pre-processing.

Technical Experience

- **Programming:** Python (Beginners Proficiency), Bash Scripting (Proficient), AI-assisted scientific coding and workflow development using various LLM platforms.
- **Quantum Chemistry:** Gaussian09, Gaussian16, ORCA, GAMESS, PSI4, Multiwfn.
- **Computational Biology:** GROMACS, g_mmpbsa, AMBER, CHARMM-GUI, AutoDock, Schrodinger GLIDE, and several online web-servers.
- **Visualization Packages:** PyMol, Chimera, VMD, Discovery Studio, Chem Craft, Mercury, Maestro, LigPlot+, MS PowerPoint.
- **Data Plotting Tools:** MS Excel, ORIGIN, XM-Grace.
- **Data Handling:** vi-Editor (Proficient), MS Excel.
- **Operating Systems:** Windows, Linux (CentOS, Ubuntu).
- **Databases:** PDB, MPDS, PubChem, Drug Bank, Uniprot, Kegg.
- **Electric Field:** TITAN.

Honours and Awards

- **Postdoctoral Fellowship (Oct 2023 - onwards):** Awarded a postdoctoral research fellowship by The Hebrew University of Jerusalem, Israel.
- **UGC PhD Fellowship (Sep 2016 - Aug 2020):** Awarded a research fellowship for four years for pursuing Ph.D. by the University Grants Commission, Govt. of India, New Delhi, India.
- **Best Poster Award (2019):** International Symposium on Advances in Functional & Biological Materials (ISAFBM-2019) organized by University of Lucknow & Humboldt Academy.
- **Graduate Aptitude Test in Engineering (2019):** Qualified GATE-2019 in Physics with AIR-1907 conducted by Indian Institute of Technology, Madras, India.

- **Joint Admission Test in M.Sc. (2014):** Qualified JAM-2014 in Physics with AIR-1831, conducted by Indian Institute of Technology, Kanpur, India.
- **College Topper (2014):** Topper in B.Sc. (Mathematics and Physics), DAV College, DDU Gorakhpur University, Gorakhpur, U.P., India.

Key Responsibilities Held

- Established and managed a fully functional computational research lab by reviving some high-performance computers at Advanced Instrumentation Research Facility, JNU, New Delhi (2020).
- Responsible for the maintenance of HPC labs, workstations and servers.
- Installed several computational chemistry and computational biology software packages; manage basic network troubleshooting, and provided user support.
- Assisted and mentored several junior graduate and dissertation students in computational quantum chemistry, computational biology, computer aided drug design, and non-covalent interactions-based projects.
- Resource person for lectures and talks for various academic courses and training programs.
- Resource person for hands-on sessions in various training programs and workshops.
- Collaborated in a non-academic industrial research and headed a research project.
- Reviewed research manuscripts, and projects reports.

Other Academic/Non-Academic Activities

- **Industry Project (NOVARTIS):** Studies involving drug delivery via molecular cavitands; structure and dynamics of cavitand-drug complexes; quantum mechanical and electronic structure calculations; molecular dynamics simulations; data collection, curation, and pre-processing for the development of machine learning model for predicting the binding affinity of cavitand-drug complexes.
- **Drug Discovery Hackathon (2022):** Preparation of course modules, delivering lectures on molecular docking and simulation, and took hands-on sessions on computational chemistry and biology tools and techniques; guiding students for project work involving data collection, curation, pre-processing and model development. A joint initiative of AICTE, CSIR and supported by Office of Principal Scientific Advisor, Govt. of India, NIC and MyGov.
- **ACDS Lecture Series:** Preparation of course study material on artificial intelligence and machine learning; delivering lectures; assessment of progress of participants; delivering hands-on sessions on computational biology and computational chemistry tools.
- **DST Student Training Program:** Preparation of course module; delivering lectures on basic mathematics and statistics for python programming; and assessment of progress of participants.

List of Publications

1. B. Panthi., **A. Pandey**, I. S. Ram, S. Begum, S. Kumar; In-silico studies of anti-tuberculosis drugs using advanced computational technique: A DFT, molecular docking, and QSAR analysis, (2025).

2. **A. Pandey**, R. R. Lenin, S. Patiyal, P. Agrawal; High Throughput Meta-analysis of Antimicrobial Peptides for Characterizing Class Specific Therapeutic Candidates: An In-Silico Approach, *Probiotics & Antimicrobial Proteins* (2025).
3. A. Adhikari, **A. Pandey***; Discerning Potent CSF-1R inhibitors for Targeting and Therapy of Neuroinflammation using Computational Approaches, *Journal of Biomolecular Structure and Dynamics*, (2024), 1-12.
4. **A. Pandey***, A. Adhikari; Modelling DNA-ligand interactions through variable force-field based MD simulations, *Indian Journal of Chemistry*, 2024, 63, 882-893.
5. **A. Pandey***, N. Kumar; Tracing the Transition from Covalent to Non-covalent Functionalization of Pyrene through C-, N-, and O- based Ionic and Radical Substrates using Quantum Mechanical Calculations, *RSC Advances*, 2023, 13, 14119-14130.
6. Y. B. Kumar, **A. Pandey**, N. Kumar, G. N. Sastry; Binding propensity and selectivity of cationic, anionic, and neutral guests with model hydrophobic hosts: A first principles study; *J. Comput. Chem.*; 2023, 44, 432.
7. S. Sikarwar, **A. Pandey**, A. Singh, B.C. Yadav, I.E. Uflyand, G.I. Dzhardimalieva; Synthesis of La₂O₃-Cr₂O₃-SrO nanocomposite by pyrolysis of metal carboxylates; its characterization, DFT studies and significance in humidity sensing; *Materials Science and Engineering: B*, 2022, 283, 115813.
8. A. Adhikari, **A. Pandey**, D. Kumar, A. K. Tiwari; Determination of Hybrid TSPO Ligands with Minimal Impact of SNP (rs6971) through Molecular Docking and MD Simulation Study; *Letters in Drug Design & Discovery*; 2022, 19, 549.
9. P. Chaudhary, D. K. Maurya, **A. Pandey**, A. Verma, R. K. Tripathi, S. Kumar, B. C. Yadav; Design and development of flexible humidity sensor for baby diaper alarm: Experimental and theoretical study; *Sensors and Actuators B: Chemical*; 2022, 350, 130818.
10. A. K. Shukla, M. K. Shrivastava, **A. Pandey**, J. Pandey; Synthesis, in vitro and computational studies of novel glycosyl-1, 2, 3-1H-triazolyl methyl benzamide derivatives as potential α -glucosidase inhibitory activity; *Bioorganic Chemistry*; 2021, 109, 104687.
11. P. Sinha, **A. Pandey**, A. K. Yadav; Vibrational Analysis and Molecular Docking Studies on some Ribonuclease-H HIV Inhibitors; *Biointerface Research in Applied Chemistry*; 2021, 11, 12796.
12. **A. Pandey**, A. Upadhyaya, S. Kumar, A. K. Yadav; Interaction, Dynamics and Stability Analysis of Some Minor Groove Binders with B-DNA Dodecamer 5'-(CGCAAATTTGCG)-3'; *Drug Des.*; 2020, 10, 172.
13. P. Chaudhary, D. K. Maurya, S. Yadav, **A. Pandey**, R. K. Tripathi, B. C. Yadav; Ultrafast response humidity sensor based on roasted gram derived carbon quantum dots: Experimental and theoretical study. *Sensors and Actuators B: Chemical*; 2020, 329, 129116.
14. **A. Pandey**, M. Misra, A. K. Yadav; Molecular Docking Studies on Binding Specificity of 3,6- and 2,7-Carbazoles With DNA Duplexes. *European Journal of Biological Research*; 2020, 11, 14.
15. **A. Pandey**, A. K. Yadav, R. Mishra, D. Kumar; Understanding interactions of DNA minor groove binders using advanced computational techniques; *The International Journal of Analytical And Experimental Modal Analysis*; 2020, 12, 1300.

16. **A. Pandey**, R. Yadav, A. Shukla, A.K. Yadav; Unveiling the Antimicrobial Activities of Dicationic Carbazoles and Related Analogs Through Computational Docking; Advanced Science, Engineering and Medicine; 2020, 1, 40.
17. R. Yadav, **A. Pandey**, N. Awasthi, A. Shukla; Molecular Docking Studies of Enzyme Binding Drugs on Family of Cytochrome P450 Enzymes; Advanced Science, Engineering and Medicine; 2020, 1, 83.
18. **A. Pandey**, R. Mishra, A. Shukla, A.K. Yadav, D. Kumar; In-silico docking studies of 2,5-bis(4- amidinophenyl) furan and its derivatives, published in Conference Proceedings of ISAFBM-2019.
19. A. Shukla, R. Mishra, **A. Pandey**, A.K. Dwivedi, D. Kumar; Interactions of Flavonols with DNA: Molecular Docking Studies, published in Conference Proceedings of ISAFBM-2019.
20. R. Shastri, N. Awasthi, D. Kumar, A. K. Yadav, D. Roy, S. P. Goutam, **A. Pandey**; A density functional theory study on structural stability and electronic properties of Co_xO_y ($X+Y=4-12$) nanoclusters; 2018, 10, 7.

Book Chapter(s)

1. **A. Pandey**, S. Hazarika, Computational work on Designing of Membrane Materials useful for Petroleum Industry. CRC Press, Taylor & Francis, 2023.
2. **A. Pandey**, N. Kumar, Y. B. Kumar, G. N. Sastry; Unravelling the factors controlling Molecular Aggregation: Examining various Non-Covalent Interactions and their Cooperativity, Elsevier, 2022 (Communicated).
3. **A. Pandey**, G. Gayathri; Free Energy Calculations and Molecular Dynamics in Ligand-Target Interactions. Bentham Science, 2026 (to be communicated soon).

Book(s)

1. A.K. Gautam, P. Kumar, V.K. Vishwakarma, **A. Pandey**, S. Mishra; Molecular Targets and Pharmacotherapy in Hepatocellular Carcinoma. Bentham Science, 2026.
2. **A. Pandey**, N. Kumar, Y.B. Kumar; Advances in Computational Molecular Sciences: From Non-Covalent Interactions to Artificial Intelligence, Bentham Science, 2026, (Proposed).

Manuscripts Communicated/Under Preparation

1. M. Taha, **A. Pandey**, A. Shurki, E. Y. Tshuva; Ti(IV) Asymmetric Acetyl-substituted Phenolato Hexacoordinate Complexes as Anticancer Chemotherapy of Enhanced Bioavailability (Communicated).
2. **A. Pandey**, D. Danovich, A. Shurki; A simpler approach to study bonding in Large Molecular Systems (under preparation).
3. **A. Pandey**, Y. B. Kumar, G. N. Sastry; Capping of Host-Guest Complexes and Effect of Cooperativity (to be communicated soon).
4. **A. Pandey**, G. Gayathri, G.N. Sastry; Size Dependency of Reactive Substrates on Covalent Functionalization of Carbon Nanostructures (under preparation).

5. **A. Pandey**, A. Adhikari, A. K. Yadav; Modelling Studies on novel Hepatitis C Virus Inhibitors: A Non-covalent & Free Energy Perspective (to be communicated soon).
6. **A. Pandey**, J. Deb, M. Paul, L. Saikia, P. K. Chattaraj; Cation- π Interactions in Biphenylene Nanoflakes: Insights from First-Principles Calculations (to be communicated soon).

Conferences/Workshops Attended and Talks Delivered

1. Awarded **Best Poster Presentation** for “In-silico studies on 2,5-bis(4-amidinophenyl) furan and its derivatives” in International Symposium on Advances in Functional & Biological Materials (ISAFBM-2019) organized by University of Lucknow & Humboldt Academy Lucknow.
2. Delivered **Oral Presentation** on “A comparative analysis of Molecular Dynamics Trajectories for amber03 & charmm27 force fields” in ISCBSC-RSG India Symposium 2019 on: Computational Biology: Connecting Domains of Science, held in, Jawaharlal Nehru University, New Delhi.
3. Delivered **two hands-on sessions** on “Using Python scripts in Data Mining” in International Workshop on: Tools and Techniques to Perform Molecular Modelling and Computer Aided Drug Design (MMTT-2021) organized by National Institute of Pharmaceutical Education & Research Guwahati, Assam, India, held from 11th to 17th Jan 2021. <https://www.youtube.com/watch?v=4PaaRBkE3hU> [4.19.00–4.54.19], <https://www.youtube.com/watch?v=cBQTvmvhrTg> [4.44.30–5.44.35].
4. Delivered a **hands-on session** on “Molecular Docking & Dynamics studies using Autodock & GROMACS” in International Workshop on: Tools and Techniques to Perform Molecular Modelling and Computer Aided Drug Design (MMTT-2021) organized by National Institute of Pharmaceutical Education & Research Guwahati, Assam, India, held from 11th to 17th Jan 2021. <https://www.youtube.com/watch?v=l3Pw1VaT2OA> [8.11.18 – 9.20.20]
5. **Presented Poster** on “Non-covalent Interactions” in North Eastern Research Conclave (NERC- 2022), held in Indian Institute of Technology, Guwahati, Assam, India.
6. Attended **7-days Training Programme** on "Machine Learning Algorithms using Python for Beginners" organized by Department of Computer Science, Assam University, Silchar-788011, Assam, India under Project Management Unit NIT Agartala; Synergistic Training Program Utilizing the Scientific and Technological Infrastructure (STUTI) an initiative of Department of Science and Technology (DST), Government of India from 16th to 22nd Jan 2023.
7. Delivered an **Invited Talk** in DAVPG College, Gorakhpur on “Computer Aided Drug Discovery” on 13th Dec 2023.

Working Genre

- Quick learner and can quickly adapt to new environments.
- Comfortable in working in a multidisciplinary institution/research organisation, and working in remote/hybrid mode.
- Strong independent work style and excellent teamwork skills.
- Well-organized and Excellent communication skills.

Reviewer For

1. Journal of Biomolecular Structure & Dynamics (Taylor and Francis) [16]
2. Scientific Reports (Springer Nature) [03]
3. Journal of Physical Chemistry A (ACS) [01]
4. Journal of Molecular Liquids (Elsevier) [02]
5. Journal of Physics Condensed Matter (IOP Publishing) [01]
6. Physical Biology (IOP Publishing) [01]
7. Medicinal Chemistry Research (Springer) [01]
8. Physical Chemistry Chemical Physics (RSC) [01]
9. In Silico Pharmacology (Springer Nature) [02]

Editing Experiences

1. Co-editor for Frontiers in Molecular Biosciences “*Advancing Protein and Peptide Research through Multiscale Modelling and Machine Learning*”

Link of Profiles

1. SCOPUS: <https://www.scopus.com/authid/detail.uri?authorId=57219981694>
2. Google Scholar: <https://scholar.google.co.in/citations?user=mIyCMpQAAAAJ&hl=en>
3. LinkedIn: <https://www.linkedin.com/in/anwesh-pandey-61831a2a5/>

References

- 1. Prof. G. Narahari Sastry**
Department of Biotechnology,
Indian Institute of Technology, Hyderabad, Telangana, India.
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- 2. Dr. Anil Kumar Yadav**
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- 3. Dr. N. Arul Murugan**
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(Anwesh Pandey)